

PCB Design using eSim

Task 7 — eSim Semester Long Internship, Autumn 2026

Regulated 5V Power Supply Board with LED Indicator

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Tool used: eSim Version 2.5 (KiCad-based)

Project files: PowerSupplyBoardDem

1. Project Overview

This is my first PCB design project, created as part of learning the complete PCB design workflow in eSim. I chose a regulated power supply circuit because it is a simple, well-understood circuit that allowed me to focus on properly learning each stage of the process — schematic capture, footprint assignment, PCB layout, multi-layer routing, and design rule checking — without the added complexity of a large or unfamiliar circuit.

The board takes an unregulated DC input, filters it, regulates it down to a stable 5V DC output using a linear voltage regulator, and includes an LED indicator that lights up to confirm the output is active. The complete design was carried out using eSim Version 2.5, following the mandatory workflow: schematic in eSchema, footprint assignment, Sch-to-PCB update, routing, DRC verification, and 3D view generation.

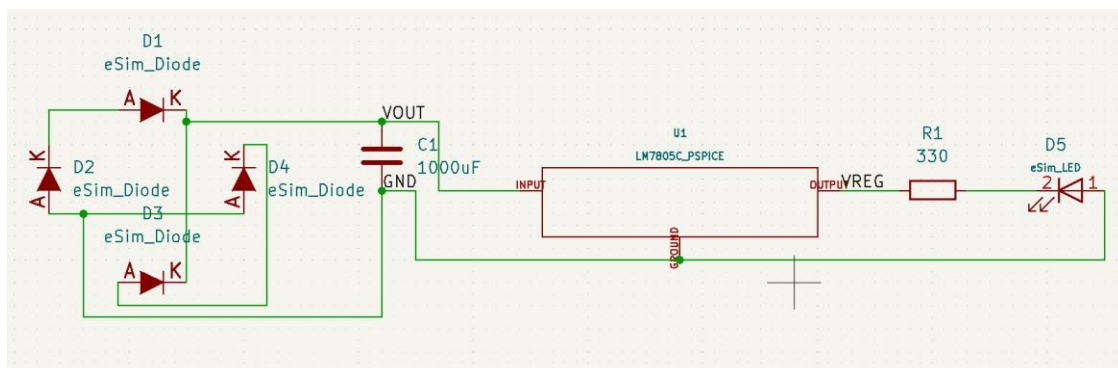
2. Circuit Description

The circuit consists of the following stages:

- Rectification/Protection (D1–D4 – eSim_Diode): Diodes placed ahead of the filtering stage to condition the incoming supply before regulation.
- Input Filtering (C1 – 1000 μ F electrolytic capacitor): Smooths the incoming DC voltage and reduces ripple before it reaches the regulator.
- Voltage Regulation (U1 – LM7805, modeled as LM7805C_PSPICE): Regulates the filtered input down to a stable, fixed 5V DC output.
- Current-Limiting Resistor (R1 – 330 Ω): Limits the current through the LED to protect it from damage.
- LED Indicator (D5 – eSim_LED): Lights up when the regulated 5V output is present, giving a simple visual confirmation that the supply is functioning correctly.

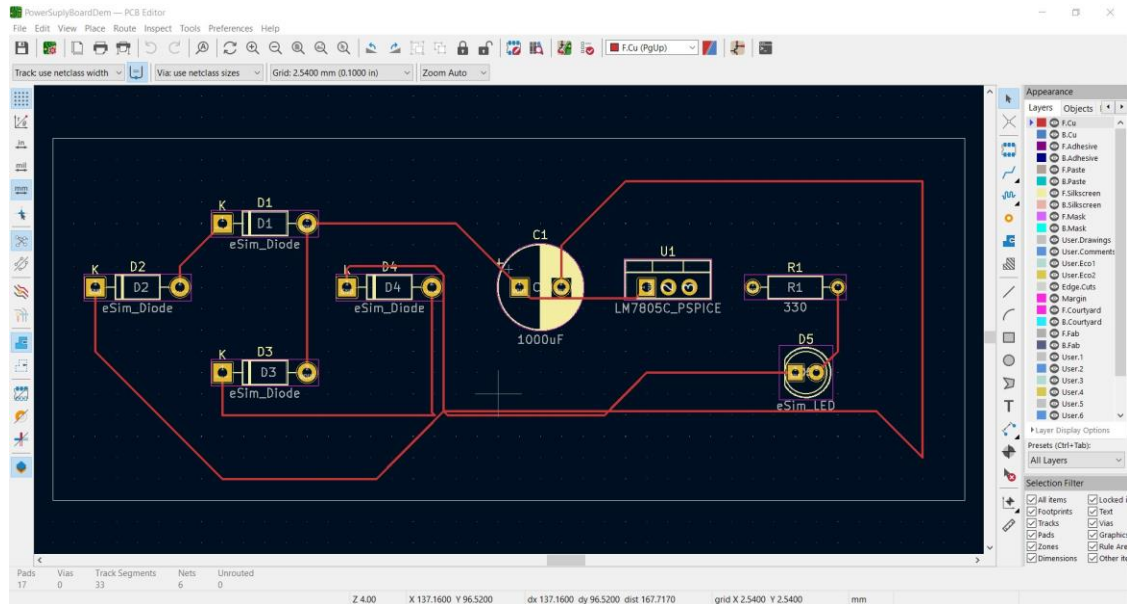
3. Schematic Screenshot

The schematic was created in eSchema (eSim's schematic editor). It shows the diode stage (D1–D4), the filtering capacitor (C1), the LM7805 voltage regulator (U1), the current-limiting resistor (R1), and the LED indicator (D5), all connected as described above.



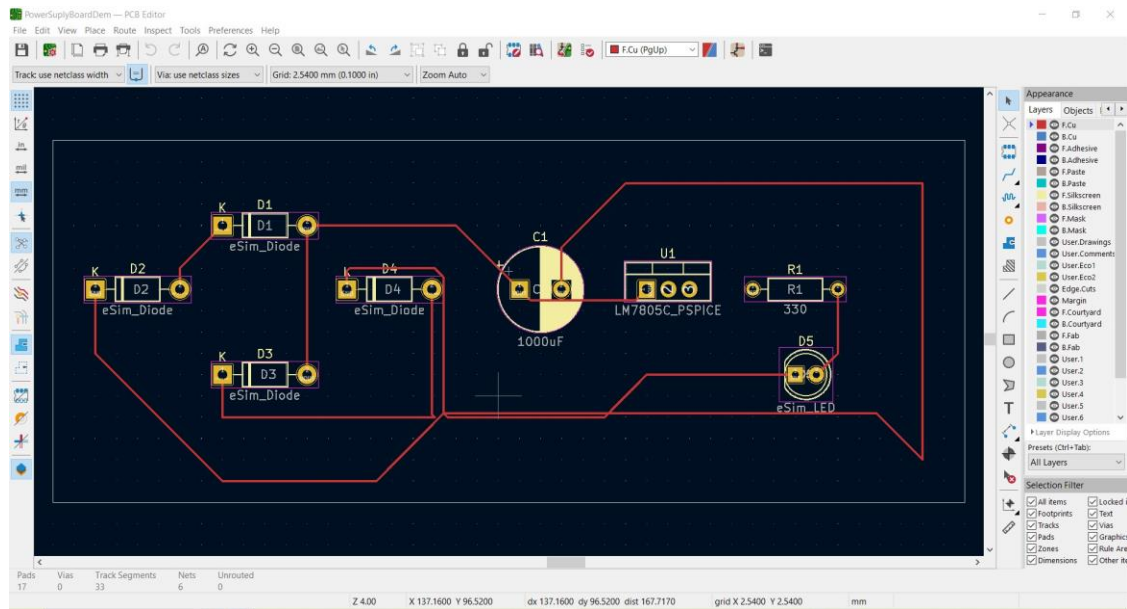
4. Footprint Assignment

Standard through-hole (THT) footprints were assigned to each component in the schematic to match real, commonly available parts: radial electrolytic capacitor footprint for C1, a 3-pin TO-220-style footprint for the LM7805 regulator (U1), axial footprints for the diodes (D1–D4), a standard resistor footprint for R1, and an LED footprint for D5. After assigning the footprints in the schematic, the PCB was generated using the Sch to PCB Update functionality in eSim, which transferred all footprints and net connections directly into the PCB editor.



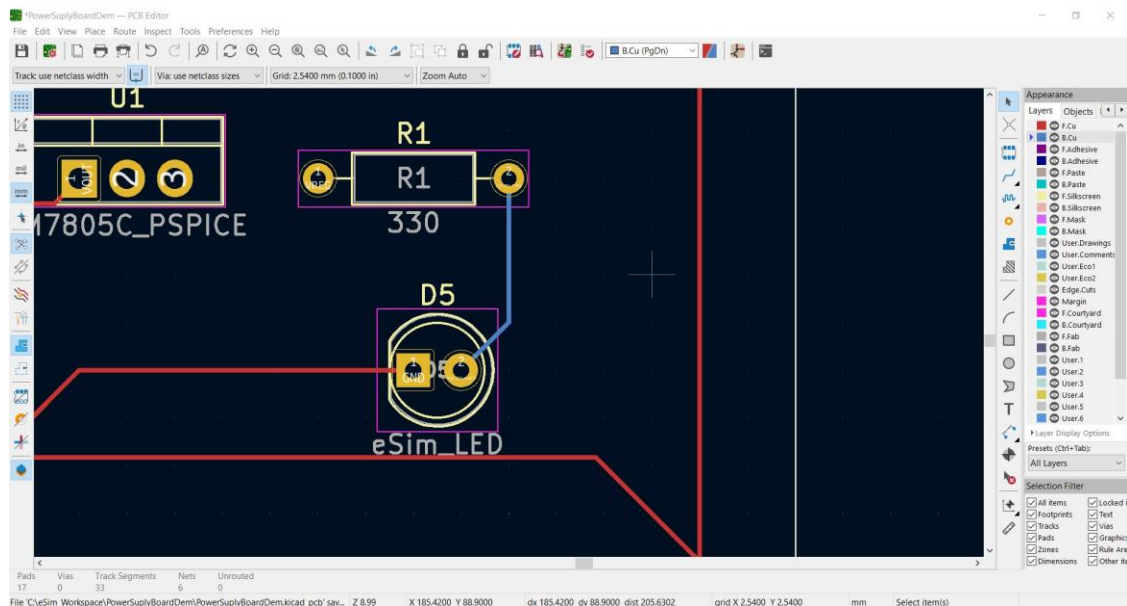
5. PCB Layout

Components were arranged on the board with adequate spacing for clean routing and readability. The board outline (Edge.Cuts) was drawn as a rectangle enclosing all components with margin on every side, satisfying the physical board-shape requirement.



6. Routing Methodology

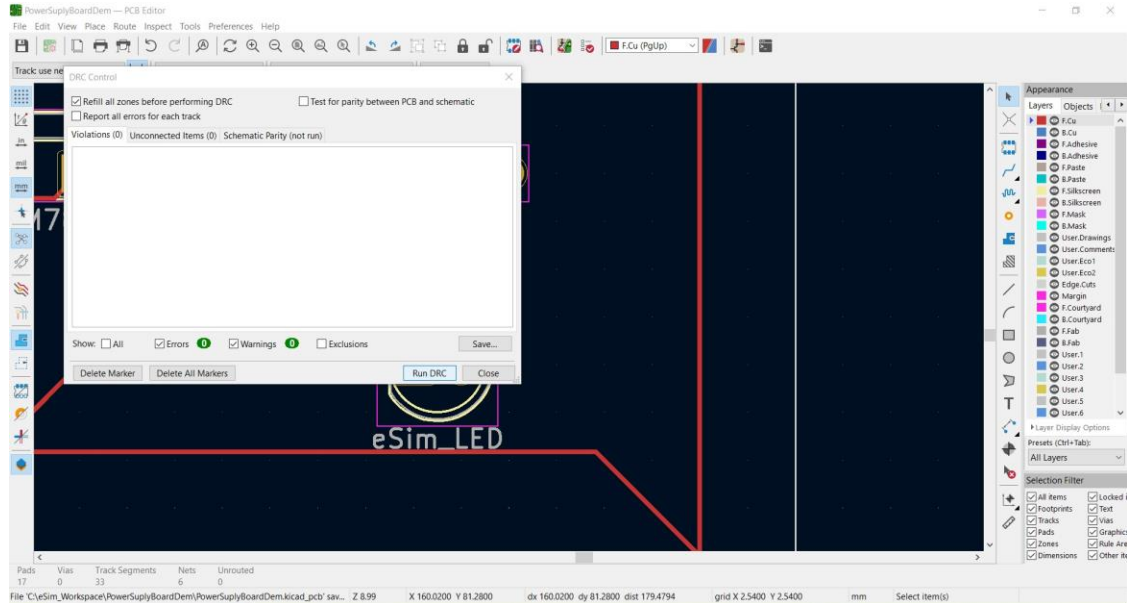
All nets were routed on the front copper layer (F.Cu) initially. To meet the requirement of using at least two copper layers, the connection between R1 and D5 was re-routed on the back copper layer (B.Cu). Since all components use through-hole (THT) footprints, their pads are inherently connected across both copper layers, allowing the back-layer trace to connect correctly without requiring a via.



The final board uses 33 routed track segments across 6 nets, with 0 unrouted connections.

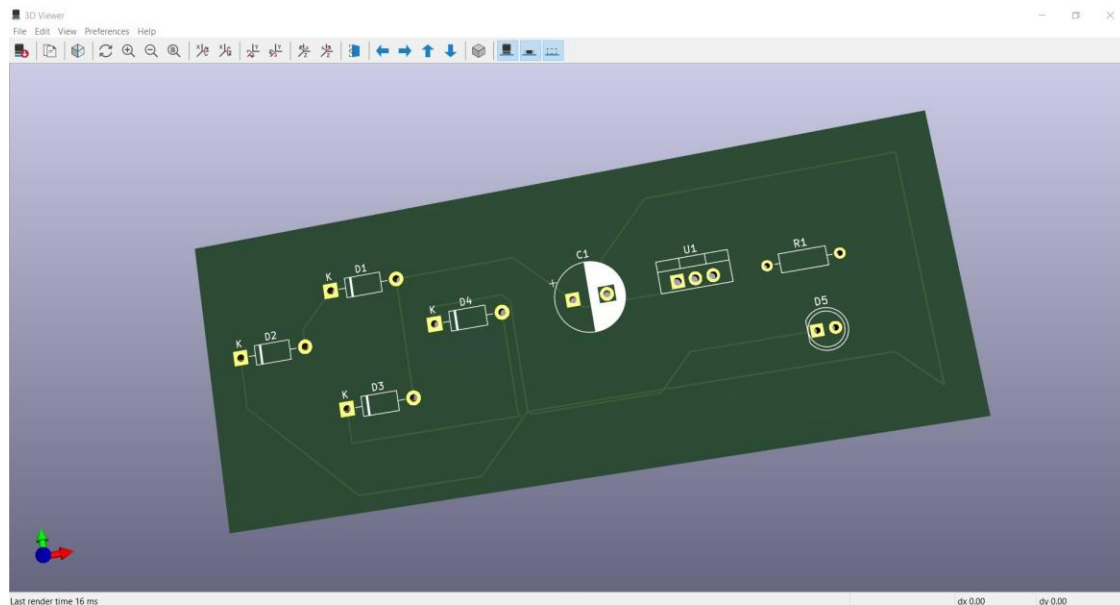
7. Design Rule Check (DRC) Results

The Design Rules Checker was run in KiCad's PCB Editor after completing the routing and drawing the board outline. The final DRC run reported 0 errors and 0 warnings, confirming the layout meets all clearance, connectivity, and board-outline requirements.



8. 3D View

The completed board was verified visually using KiCad's built-in 3D Viewer, accessible from eSim's PCB Editor (View → 3D Viewer). The render confirms correct placement of all components, the green solder mask, the board outline, and the routed copper traces.



9. Design Considerations and Conclusions

- The circuit was intentionally kept simple as this was my first PCB design, allowing me to focus on learning the complete eSim/KiCad workflow correctly rather than managing circuit complexity.
- Through-hole footprints were chosen throughout, which simplified two-layer routing since THT pads connect across both copper layers without needing vias.
- The board outline was drawn with margin around all components to avoid tight clearances near the board edge.
- Design Rule Check was run repeatedly throughout the process (not just at the end) to catch and fix issues, such as an orphaned trace, as early as possible.
- The final design meets all mandatory Task 7 requirements: schematic capture, footprint assignment, Sch-to-PCB update, complete routing across two copper layers, a clean DRC run, and a verified 3D view.

Overall, this project gave me hands-on experience with the end-to-end PCB design process in eSim, from circuit schematic to a fully routed, DRC-clean, two-layer board ready for fabrication. I plan to build on this foundation with more complex designs going forward.